From Usage to Retained Demand: A Structural Audit Framework for Institutionally Connected Digital Assets

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Abstract

This paper proposes a structural audit framework for institutionally connected digital assets. The central claim is not that such assets must dominate payment flows, nor that usage alone is sufficient to explain their economic significance. Rather, the relevant question is whether institutional expansion creates conditions under which retaining a digital asset becomes operationally necessary in inventory, collateral, liquidity-buffer, or waiting-liquidity form.

The paper argues that transaction flow should not be conflated with retained demand, and that retained demand itself should be decomposed into inventory demand, collateral demand, liquidity-buffer demand, and waiting liquidity, rather than being reduced to a single notion of "usage." It also separates domain expansion, asset selection, and indispensability, and distinguishes the commercial success of surrounding infrastructure from the retained demand of the asset itself.

Although XRP functions as the primary motivating case, the contribution of this paper is not to offer an XRP-specific price thesis. It is to provide a conditional audit framework for evaluating whether institutionally connected digital assets generate retained demand under specific structural conditions. The framework is explicitly falsifiable: it can fail if bridge-like overflow does not emerge, if the asset is not selected, or if surrounding institutional stacks expand while bypassing the asset itself.

1. Introduction

A large share of digital asset analysis remains centered on visible usage. Analysts typically ask whether an asset is being used, whether transaction volume is growing, whether adoption headlines are accumulating, or whether surrounding infrastructure appears to be expanding. While such observations are not irrelevant, they do not by themselves answer a more structurally important question: under what conditions does a digital asset become economically necessary to retain?

This paper argues that the distinction between usage and retained demand is analytically decisive for institutionally connected digital assets. A settlement-connected asset may appear in transactional flow without becoming economically important in inventory form, and conversely, an asset may acquire structural significance because systems begin to require it in inventory, collateral, liquidity-buffer, or waiting-liquidity form. The relevant question, therefore, is not merely whether an asset is used, but whether the surrounding institutional environment creates conditions under which it must be held.

This distinction matters because the next phase of digital financial infrastructure is increasingly shaped by institutional embedding rather than isolated token usage. Stablecoins, tokenized deposits, shared ledgers,

tokenized treasuries, custody platforms, and integrated treasury systems are expanding the range of domains in which digitally connected assets may potentially operate. Yet domain expansion does not by itself imply that any particular asset has been selected, and selection does not by itself imply indispensability. These stages must be separated analytically if one is to avoid conflating infrastructure growth with asset-specific retained demand.

The paper is motivated primarily by the XRP case, but its contribution is methodological rather than directional. The framework does not assume that the commercial success of Ripple, RLUSD, XRPL, or adjacent infrastructure automatically translates into asset-specific demand. Instead, it asks whether institutional expansion generates retained demand for the asset itself, or whether the surrounding stack scales while bypassing it. This is not a secondary caveat but a central audit discipline: the framework is designed to identify the conditions under which retained demand emerges, and the conditions under which that thesis fails.[1]

The contribution of this paper is fourfold. First, it proposes retained demand as a central audit category for institutionally connected digital assets. Second, it decomposes demand into transaction flow, inventory demand, collateral demand, liquidity buffers, and waiting liquidity, instead of treating "usage" as a sufficient catch-all category. Third, it separates domain expansion, asset selection, and asset indispensability as distinct analytical layers. Fourth, it embeds these distinctions within a falsifiable structure that explicitly includes bypass risk, overflow failure, and non-selection as invalidation pathways.

This paper therefore defines the first layer of the retained-demand audit framework: retained demand by type. Later papers extend this base framework to settlement-stack closure, required liquidity density, stack-level compression and bypass, evidence-gated sizing, and the location of retained demand after user abstraction. The present paper does not attempt to resolve those later questions; it establishes the conceptual instrument through which they can be asked more precisely.

The remainder of the paper proceeds as follows. Section 2 explains why usage is not an adequate stand-alone category for evaluating institutionally connected digital assets. Section 3 develops retained demand as an audit category and clarifies its internal components. Section 4 separates expansion, selection, and indispensability. Section 5 examines why infrastructure success does not necessarily imply asset demand, with particular attention to bypass risk. Section 6 presents the main falsification conditions. Section 7 briefly situates XRP as the primary motivating case. Section 8 concludes.

2. Why Usage Is Not Enough

A recurring weakness in digital asset analysis is the tendency to treat usage as a sufficient proxy for economic significance. Transaction count, transfer volume, network activity, integration announcements, and payment-related headlines are often taken as evidence that a given asset is becoming more valuable or more structurally important. But visible flow is not the same thing as retained economic necessity. An asset can appear repeatedly in transactions while remaining operationally substitutable, balance-sheet light, or strategically unnecessary to hold.

This gap matters most in institutionally connected settings. Once an asset is assessed not merely as a retail token but as a potential component of treasury workflows, collateral structures, settlement routing, or cross-ledger interoperability, the relevant question changes. What matters is no longer only whether the

asset passes through the system, but whether the system begins to rely on it in held form. In other words, the difference between flow and retention becomes analytically central.

The problem with usage-centered analysis is not that usage is irrelevant. It is that usage collapses several distinct categories into one observational surface. A transfer may reflect temporary routing, speculative activity, arbitrage, collateral movement, operational prefunding, market-making inventory adjustment, or end-user payment demand. These are not interchangeable. Some imply balance-sheet commitment; others do not. Some generate retained demand; others merely generate momentary appearance in the flow. To infer structural importance from usage without separating these mechanisms is to risk category error.

A related difficulty is that infrastructure growth is often mistaken for asset demand. When new custody platforms, treasury integrations, tokenized-cash systems, payment rails, or stablecoin networks emerge, observers frequently assume that the digital asset associated with the surrounding ecosystem necessarily benefits. But this assumption does not hold in general. Infrastructure can expand while reducing, minimizing, or bypassing the role of the asset in question. Indeed, in some cases improved infrastructure may increase the efficiency of closed-loop settlement precisely by decreasing the need for an external connective asset.

For this reason, institutionally connected digital assets cannot be evaluated only through visible use or ecosystem growth. They must be audited through a second-order question: does expansion create conditions under which the asset must be retained in some economically meaningful form? If the answer is no, then high usage or high infrastructure visibility may still coexist with weak retained demand. If the answer is yes, then the asset may acquire economic relevance that is not immediately obvious from transaction flow alone.

This paper therefore treats usage as an incomplete observational category. It may serve as an input, signal, or contextual clue, but not as a stand-alone explanation of economic significance. The core audit problem is not whether the asset appears in motion, but whether institutional frictions, timing mismatches, collateral requirements, or liquidity needs generate reasons to hold it. The difference is decisive, because it marks the line between an asset that is merely present in the system and an asset that the system begins to require.

3. Retained Demand as an Audit Category

If usage is an incomplete category, what should replace it as the central audit lens? This paper proposes retained demand as the more relevant category for institutionally connected digital assets. Retained demand refers not to directional conviction or speculative preference, but to situations in which an asset must be held, reserved, posted, buffered, or maintained in order for a workflow, settlement process, collateral structure, or liquidity arrangement to function with sufficient speed, resilience, or certainty.[2]

The advantage of this category is that it shifts the analysis from surface flow to operational necessity. Instead of asking only whether an asset is circulating, one asks whether the surrounding system creates a reason to keep the asset on hand. Such reasons may arise from timing gaps, interoperability frictions, collateral requirements, market-making obligations, treasury management, prefunding minimization, volatility management, or stress-period liquidity needs. The common feature is that the asset is not merely passing through the system; it is being retained because not holding it would introduce cost, delay, risk, or

fragility. This concern is not foreign to existing payment, liquidity, and reserve-management literature, which has long recognised that operationally significant holdings cannot be inferred from visible transactional flow alone (see, e.g., BCBS 2013; Testi et al. 2023).

To make this category analytically useful, retained demand must be decomposed. In this framework, four components are especially important.

First, inventory demand refers to the need to maintain the asset as operational inventory within a workflow. This may include market-making inventory, settlement inventory, routing inventory, or treasury-held balances intended to support continuous functionality. The relevant issue is not whether the asset is eventually used, but whether it must be present in advance or held across time in order to prevent disruption. Inventory demand is therefore tied to operational continuity and workflow support, rather than to precautionary shock absorption as such. Inventory demand is tied to anticipated recurring operational use, whereas liquidity buffer demand is maintained primarily against mismatch, stress, or execution uncertainty.

Second, collateral demand refers to cases in which the asset is posted, pledged, reserved, or recognized as support for obligations, exposures, or liquidity commitments. This category is distinct from transaction use because collateral operates through lockup, eligibility, and balance-sheet treatment rather than through mere transfer. It is also distinct from ordinary operational holding: the asset here is retained because it supports claims, exposures, or margin-like requirements, not simply because it must remain available for routing or execution. If an asset becomes acceptable or useful in collateral form, its economic role changes materially.

Third, liquidity buffer demand refers to balances held not because they are in constant motion, but because they serve as shock absorbers against uncertainty, mismatch, volatility, or execution risk. Institutions often hold buffers precisely because smooth systems still require protection against discontinuity. Unlike operational inventory, buffer demand is precautionary in orientation: it exists to absorb stress, mismatch, or execution slippage rather than to sustain ordinary workflow continuity. If a digital asset becomes part of that protective layer, its importance cannot be inferred from visible transactional flow alone.

Fourth, waiting liquidity refers to balances held across temporal gaps while a transfer, conversion, settlement sequence, or cross-domain process is pending, staged, or conditionally linked. This category matters because multi-step institutional workflows are rarely instantaneous in a strong operational sense, even when individual components are digitally accelerated. Time still exists inside the system, and where time exists, capital often waits. Waiting liquidity is therefore temporally staged rather than primarily precautionary: it is retained because the process is not yet complete, not simply because the institution wishes to hold an extra safety margin.

These components overlap in practice, but analytically they should not be collapsed into a single notion of use. The purpose of decomposition is not terminological complexity for its own sake. It is to clarify that economically relevant demand can arise through several channels that are invisible, understated, or misread when the analyst focuses only on movement. The more institutionally embedded the environment becomes, the more important this decomposition is likely to become.

Retained demand does not imply that any specific asset will necessarily succeed. It is an audit category, not a conclusion. The framework proposed here does not begin by assuming that an asset deserves to be held. It asks what structural conditions would create such a requirement, and what evidence would show that the requirement is not emerging. In that sense, retained demand is best understood not as a bullish reinterpretation of usage, but as a stricter and more falsifiable standard for evaluating economic relevance.

4. Separating Expansion, Selection, and Indispensability

A central source of confusion in digital asset analysis is the tendency to compress distinct stages of structural development into a single narrative of progress. Once an ecosystem expands, many observers infer that the associated asset is therefore gaining traction; once it gains traction, they further infer that it is becoming indispensable. These inferences are often made implicitly, but they should not be treated as analytically interchangeable. This paper therefore separates three layers that are frequently conflated: domain expansion, asset selection, and asset indispensability.

Domain expansion refers to the growth of institutional, regulatory, technological, or operational domains in which a digital asset could potentially participate. This includes the emergence of tokenized cash systems, cross-ledger workflows, treasury integration, custody infrastructure, settlement modernization, programmable collateral arrangements, and interoperable market structures. Domain expansion enlarges the addressable space. It tells us that new arenas are opening. But it does not tell us which asset, if any, will occupy a meaningful role within them.

Asset selection refers to the next step: whether a given asset is actually chosen within one or more of those domains. Selection is stronger than possibility, but still weaker than indispensability. An asset may be selected for routing, bridging, collateral support, inventory management, or temporary operational use without becoming difficult to replace. Selection indicates realized participation, not structural necessity.

Asset indispensability refers to the strongest layer. It arises when removing the asset materially worsens cost, resilience, timing, slippage, capital efficiency, or interoperability relative to available alternatives. Indispensability does not require monopoly. An asset may remain one option among several while still occupying a role whose removal degrades the system in a non-trivial way. What matters is not exclusivity but functional consequence.

The distinction among these three layers is essential because each stage requires different evidence. Domain expansion can often be observed through regulatory change, infrastructure buildout, product launches, or institutional experimentation. Selection requires more specific evidence that an asset is being used within a relevant operational channel. Indispensability requires an even stronger standard: evidence that the system performs materially worse without the asset, or that substitution imposes meaningful friction, cost, or fragility.

This layered separation also protects against a common analytical error: treating ecosystem development as if it were already proof of asset demand. In practice, a domain may expand while no specific asset is selected; an asset may be selected while remaining easily replaceable; and even substantial ecosystem growth may occur in ways that bypass the asset entirely. Without separating these layers, analysts are prone to importing conclusions from one stage into another.

In the present framework, these layers are not merely descriptive. They form a monitoring sequence and a discipline of inference. Domain expansion is necessary but not sufficient. Selection is important but still not conclusive. Indispensability is the strongest condition, but also the hardest to establish. The benefit of this sequence is that it creates a more controlled pathway between observation and claim. It slows down premature conclusions while still allowing structural progress to be tracked with precision.

5. Infrastructure Success Does Not Equal Asset Demand

One of the most persistent errors in ecosystem-based digital asset analysis is the assumption that the success of surrounding infrastructure necessarily benefits the associated asset. The intuition is understandable: if a network's custody stack grows, if payment connectivity expands, if tokenized cash products gain traction, or if treasury workflows become more deeply integrated, then the asset at the center of the ecosystem is presumed to gain in relevance. But this inference is analytically unsafe. Infrastructure success and asset demand must be separated.

This distinction is especially important in institutionally connected environments, where infrastructure layers often optimize for efficiency, compliance, risk reduction, and balance-sheet control. Under those conditions, it is entirely possible for the surrounding stack to become more successful precisely by reducing the need for the asset that observers assume will benefit. Closed-loop settlement, internal netting, stablecoin expansion, tokenized deposit systems, or integrated treasury rails may all improve institutional workflows while minimizing the role of an external connective asset.

This possibility is especially important in case-motivated analysis where a digital asset is closely associated with a broader commercial or infrastructural stack. The growth of that surrounding stack may enlarge the relevant institutional domain without creating retained demand for the asset itself. One of the central risks is bypass: the stack grows, the workflows deepen, the institutional footprint expands, but inventory, collateral, liquidity-buffer, or waiting-liquidity demand for the asset does not follow. If this occurs, surrounding success should not be misread as indirect proof of asset-specific economic significance.

Bypass risk is therefore not a peripheral caveat. It is a core audit requirement. Any framework that begins with infrastructure growth and moves too quickly toward asset-specific conclusions is vulnerable to overstatement. The more successful the surrounding infrastructure becomes, the more important it is to ask whether the asset remains necessary at all. In some cases, improved infrastructure may increase the efficiency of asset-linked workflows. In other cases, it may absorb the relevant functions internally and render the asset optional or marginal.

For this reason, institutionally connected digital assets should be assessed under a stricter condition: not whether ecosystem components are succeeding, but whether that success creates conditions under which the asset itself must be retained. This is the point at which the present framework departs from both simplistic adoption narratives and direct ecosystem-value transfer assumptions. A stack can expand without generating retained demand. Conversely, if retained demand does emerge, it must be shown through evidence specific to the asset's role in inventory, collateral, liquidity support, or temporal bridging.

The analytical implication is straightforward. Ecosystem growth may matter, but it matters differently at different layers. At the level of domain expansion, surrounding infrastructure success can be highly relevant. At the level of asset demand, however, it is only indirectly informative. The burden of proof rises as the claim becomes more asset-specific. This is why the distinction between expansion, selection, and indispensability is not optional. Without it, infrastructure success is too easily mistaken for asset necessity.

6. Falsification Conditions

A useful audit framework must do more than organize supportive observations. It must also identify the conditions under which its own central thesis would fail. The present framework is therefore explicitly falsifiable. It does not claim that institutionally connected digital assets necessarily generate retained demand, nor does it assume that any specific asset will move from visibility to structural necessity. Instead, it specifies a set of pathways through which the thesis may be invalidated.

The first falsification condition is overflow failure. The framework assumes that as institutional domains expand, there may remain cross-domain, cross-ledger, or cross-structure gaps in which a connective asset could play a role. But if those gaps do not become economically meaningful—because systems close efficiently, interoperability becomes internally managed, or layered infrastructures remain sufficiently integrated without an external bridge—then the retained demand thesis weakens materially. In that case, the relevant overflow does not develop. Confirming pressure for the thesis would consist in repeated evidence that economically meaningful settlement or liquidity gaps persist across domains. Invalidating pressure would consist in convergent evidence that such gaps are routinely internalized, netted away, or closed without any need for an external connective asset. In that sense, invalidating pressure is also present where anticipated cross-domain overflow fails to become economically material despite surrounding infrastructure expansion.

The second condition is non-selection. Even if bridge-like gaps remain, the asset in question may simply not be selected. Other assets, other rails, internalized routing solutions, tokenized deposits, stablecoin systems, or institution-specific mechanisms may occupy the relevant function instead. A structural space may exist without being filled by the asset under analysis. Domain expansion is therefore not enough; selection must be demonstrated rather than assumed. Confirming pressure would consist in asset-specific evidence of actual use within the relevant institutional channel. Invalidating pressure would consist in repeated evidence that the relevant function is being filled by substitutes while the asset under analysis remains absent or marginal.

The third condition is infrastructure bypass. This is one of the most important invalidation pathways. It is possible for the surrounding stack to grow while bypassing the asset itself. Payment systems may scale, custody may deepen, treasury tools may become embedded, and settlement infrastructure may modernize without generating meaningful inventory, collateral, liquidity-buffer, or waiting-liquidity demand for the asset. If this occurs, then ecosystem success should be interpreted as evidence against the stronger asset-demand thesis rather than in favor of it. Confirming pressure would consist in evidence that ecosystem growth is accompanied by held-form demand specific to the asset. Invalidating pressure would consist in evidence that the relevant workflows scale successfully while the asset is not retained in any operationally meaningful way.

The fourth condition is stablecoin or deposit-token full coverage. If stablecoins, tokenized deposits, shared ledgers, or tightly coordinated institutional cash systems become capable of handling the relevant workflows end-to-end with sufficient efficiency, then the need for a separate connective asset may decline sharply. Under those conditions, what once appeared to be a role for an intermediate asset may instead be internalized within the expanding institutional settlement architecture. Confirming pressure would consist in persistent frictions or coverage limits that leave room for an external connective role. Invalidating pressure would consist in evidence that end-to-end coverage is sufficiently broad and efficient that no separate retained bridge function is needed.

The fifth condition is efficiency compression through netting, orchestration, or synchronization. Established payment-system literature has likewise long recognised that netting, liquidity-saving mechanisms, and related design choices can materially reduce the amount of liquidity participants must fund or hold (see, e.g., CPMI 2023; Testi et al. 2023).[3] The more effectively institutions can compress settlement needs through pre-coordination, netting, smart routing, synchronized ledgers, or AI-enhanced liquidity optimization, the smaller the operational space in which an independently retained bridge asset is necessary. If such mechanisms materially reduce the need for transitional inventory or liquidity buffers, then the retained demand thesis must be weakened accordingly. Confirming pressure would consist in evidence that residual liquidity gaps remain even after netting and orchestration improvements. Invalidating pressure would consist in evidence that such mechanisms compress the relevant need so far that retained bridge inventory ceases to be economically material. In this sense, liquidity-saving design is not merely an efficiency refinement but a direct invalidating pressure on claims of persistent asset-specific retained demand.

The sixth condition is absence of retained-demand evidence. Usage may expand. Headlines may multiply. Ecosystem narratives may strengthen. But if no corresponding evidence emerges that the asset is being held in operationally meaningful form—whether as inventory, collateral, liquidity reserve, or waiting liquidity—then the stronger structural claim has not been established. Visibility without retention is not enough. Confirming pressure would consist in credible evidence of held-form necessity at the operational, collateral, or liquidity-management level. Invalidating pressure would consist in a repeated pattern of flow visibility without any corresponding signs of asset-specific retention.

These falsification conditions are not secondary limitations added for rhetorical balance. They are part of the framework's core logic. Their purpose is to prevent category errors, to constrain overreading, and to preserve analytical discipline as institutional infrastructure evolves. A retained demand thesis that cannot specify its own failure conditions is not an audit framework; it is merely a directional narrative.

7. XRP as a Primary Stress Case

The framework developed in this paper is not limited to XRP, but XRP provides a particularly revealing motivating case because it sits at the intersection of several layers that are often conflated in digital asset analysis. It has been discussed as a settlement asset, a bridge asset, a liquidity instrument, and an infrastructure-linked token associated with a broader commercial ecosystem. Precisely for this reason, the XRP case makes visible the analytical errors that the present framework is designed to avoid.

The purpose of this section is not to conclude that XRP has already achieved structural indispensability, but to test how the proposed framework behaves when applied to a case in which domain expansion, possible selection, and bypass risk coexist.

First, XRP illustrates the importance of separating surrounding infrastructure success from asset-specific retained demand. Ripple, XRPL, RLUSD, custody systems, treasury integrations, payment services, and adjacent institutional tools may all expand in commercially significant ways. But none of these developments, by themselves, prove that XRP is being retained in economically meaningful form. The temptation to read ecosystem progress as direct evidence of XRP demand is strong; the framework proposed here treats that temptation as a category error unless additional evidence is provided.

Second, XRP is a useful case for distinguishing domain expansion from asset selection. The range of domains in which a connective digital asset could in principle participate may expand substantially: cross-border settlement, tokenized cash systems, treasury routing, collateral mobility, interoperable liquidity pathways, and institutional market infrastructure. Yet the expansion of these domains does not establish that XRP has been selected within them. Still less does it establish that XRP has become indispensable. The XRP case therefore demonstrates why layered inference is necessary.

Third, XRP helps foreground the importance of bypass risk. If surrounding institutional infrastructure becomes more integrated, more regulated, and more efficient, this may either increase or decrease the need for XRP. A framework that begins by assuming positive transfer from ecosystem development to asset demand cannot adequately evaluate this risk. The present audit structure instead requires the analyst to ask whether XRP is actually being retained in inventory, collateral, liquidity-buffer, or waiting-liquidity form, and whether removing XRP would materially worsen the relevant workflow.

This makes XRP an especially demanding case rather than an easy one. It is not useful because it straightforwardly confirms the framework, but because it forces the framework to confront all major invalidation paths at once: non-selection, bypass, internalized institutional settlement, stablecoin full-coverage, and the possibility that visible ecosystem growth may coexist with limited retained demand for the asset itself. In that sense, XRP is not merely an example attached to the theory. It is a primary stress case for the audit method.

The implication is clear. Any stronger claim about XRP—whether in relation to sizing, structural significance, or eventual indispensability—must come later, and must be built on evidence specific to retained demand rather than on ecosystem adjacency alone. That is why the present paper stops at the level of framework. It establishes the criteria under which the XRP question should be asked, not the final answer to that question.

8. Conclusion

This paper has argued that institutionally connected digital assets should not be evaluated through usage alone. Visible flow, adoption headlines, network activity, and surrounding infrastructure growth may all be relevant observations, but they do not by themselves establish economic significance. The more important question is whether institutional expansion creates conditions under which a digital asset must be retained in inventory, collateral, liquidity-buffer, or waiting-liquidity form.

From that starting point, the paper proposed a structural audit framework organized around several distinctions. It separated usage from retained demand, infrastructure success from asset demand, and domain expansion from asset selection and asset indispensability. It also argued that retained demand should be decomposed rather than treated as a vague extension of usage. Finally, it embedded these distinctions within an explicitly falsifiable structure that includes overflow failure, non-selection, bypass, full-coverage alternatives, efficiency compression, and absence of retained-demand evidence as invalidation pathways.

The purpose of this framework is not to support a directional narrative about any one asset. It is to impose a stricter standard on claims of structural significance in digitally connected settlement environments. It asks not whether an asset can appear in the flow, but whether the surrounding system begins to require it in held form.

XRP served here as the primary motivating case because it makes these analytical distinctions unusually visible. Yet the broader contribution of the paper is methodological. It offers a way to evaluate digital assets that is less dependent on adoption rhetoric, less vulnerable to ecosystem overreading, and more attentive to the balance-sheet, collateral, and liquidity structures through which retained demand may or may not emerge.

Much remains unresolved. The framework does not by itself prove that any specific asset will become indispensable. Nor does it eliminate the need for future work on settlement stack competition, conditional sizing, collateral lockup, and the empirical evidence required to establish retained demand in practice. What it does provide is a more disciplined starting point.

In that sense, this paper should be read as the conceptual starting point of a broader retained-demand audit series: it defines the demand categories that later analyses can test against settlement-stack closure, liquidity-density burden, evidence-gated sizing, and backend balance-sheet location.

If institutionally connected digital assets are to be evaluated seriously, they must be evaluated not only by what moves through them, but by what must remain inside them.

Footnotes

1. This paper is adjacent to, but not dependent on, prior public-facing structural analyses concerning settlement scale, liquidity mechanics, netting, and the possible economic role of digitally connected assets in payment or transfer systems. The present paper does not adopt any external sizing thesis as its core claim. Its distinct contribution lies instead in proposing a falsifiable audit framework centered on retained demand, infrastructure–asset separation, and explicit invalidation conditions.

2. Retained demand, as used in this paper, overlaps partially with established concerns such as precautionary balances, intraday liquidity management, collateral posting, and treasury liquidity reserves, but is not reducible to any one of them. The term is introduced as an audit concept for institutionally connected digital assets: it draws attention to the conditions under which assets must be held in advance, maintained under uncertainty, or reserved for resilience, participation, or positional readiness, even when observable usage does not by itself reveal their full economic role.

3. Existing payment-system theory already recognises that netting, offsetting, and liquidity-saving mechanisms can materially reduce the amount of liquidity that participants must mobilise or maintain during the operating day. For the present framework, this is not a secondary efficiency detail but a genuine falsification line: where system design compresses required holdings, claims about persistent asset-specific retained demand weaken accordingly.

Keywords

Retained Demand; Digital Assets; Institutional Settlement; Inventory Demand; Collateral Demand; Liquidity Buffers; Waiting Liquidity; Tokenization; Financial Market Infrastructure; Structural Audit; XRP

Author Note

This paper is presented as a working paper and should be read as a framework-oriented contribution. It is the first paper in a retained-demand audit series for institutionally connected digital assets. Its purpose is to clarify how such assets may be evaluated when visible usage, retained demand, infrastructure growth, and asset-specific necessity are treated as analytically distinct questions. XRP is used here as a motivating case and stress case, but the contribution of the paper is methodological rather than advocative. The paper does not provide investment advice, a price target, or a recommendation to buy, sell, or hold any digital asset.

Short Description

A working paper proposing a falsifiable structural audit framework for institutionally connected digital assets. The paper distinguishes visible usage from retained demand, decomposes retained demand into inventory demand, collateral demand, liquidity-buffer demand, and waiting liquidity, and separates infrastructure success from asset-specific demand. XRP is used as a motivating case and stress case, not as the subject of a price thesis.

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